

E-Commerce Performance & Customer Analytics Report

Prepared from dashboard analysis

Formal Data Analysis Report

1. Executive Summary

Executive KPIs

Metric	Value	Metric	Value
Total Sales	21.78M	Total Orders	17K
Total Profit	5.70M	Items Sold	51.34K
Total Discount	1.19M	Average Order Value	1.28K
Profit Margin	26.18%	Average Rating	3.90 / 5
Delayed Orders	32.24%	On-Time Orders	67.76%

Data Quality Summary

The dataset is clean, well-structured, and ready for analysis. No missing-data issues or obvious quality concerns were identified across univariate, categorical, or bivariate checks. Accordingly, the findings presented in this report should be interpreted as genuine business signals rather than data artifacts.

Overall Business Health

The business is fundamentally healthy, supported by solid profitability, strong customer satisfaction, and balanced product demand. However, revenue is concentrated across specific periods, categories, and cities, and operational performance is weakened by a meaningful delivery-delay issue beneath otherwise strong headline results.

Key Highlight 1: Electronics and Bengaluru together represent the business's core growth engine, with Electronics contributing over half of total revenue and Bengaluru generating roughly double the revenue of the next-highest city.

Key Highlight 2: Q1 is a major seasonal spike, with monthly revenue at approximately 3M from January to March before declining to roughly 1M per month for the rest of the year.

Key Highlight 3: Nearly 1 in 3 orders are delayed, indicating a material logistics concern despite generally acceptable overall delivery-time distribution.

2. Overall Business Overview

The available dashboard analysis indicates a business with strong top-line revenue, healthy margins, and positive customer feedback. Total sales reached 21.78M, producing 5.70M in profit and a profit margin of 26.18%. Order activity is substantial, with 17K total orders and 51.34K items sold, while the average order value stands at 1.28K.

Customer sentiment appears broadly positive, with an average rating of 3.90 out of 5 and a rating distribution heavily concentrated in the 4-star to 5-star range. At the same time, operational delivery performance requires attention, as 32.24% of all orders are delayed. The business therefore shows strong commercial performance, but with notable dependency risks and operational inefficiencies that should be addressed to support sustainable growth.

3. Key Business Drivers

A. Revenue by City

City	Approximate Revenue Contribution
Bengaluru	Highest; roughly 2x the next city
Delhi (NCR)	Second tier
Mumbai	Second tier
Hyderabad	Second tier
Chennai	Second tier
Kolkata, Pune, Ahmedabad	Mid tier
Lucknow, Jaipur	Lowest contribution

Market hierarchy: Bengaluru → Delhi (NCR) → Mumbai → Hyderabad → Chennai. Metro cities contribute the majority of total sales, while smaller cities remain comparatively underdeveloped.

B. Revenue by Product Category

Category	Revenue Contribution
Electronics	10.48M (51.63%); dominant and also the highest-profit category
Sports	4.02M (19.82%)

Fashion	3.21M (15.79%)
Home & Garden	Mid-tier contributor
Toys, Beauty, Food, Books	Long-tail categories with minimal contribution

Key fact: Electronics leads in every region, confirming that its performance is a true business strength rather than a localized anomaly.

C. Revenue by Time (Seasonality)

Monthly revenue is strongest in **Q1 (January to March, approximately 3M per month)**. After March, revenue declines sharply and stabilizes at roughly **1M per month** for the remaining nine months. This estimated 65–70% post-Q1 decline is the single largest fluctuation observed in the dataset.

D. Customer Demographics and Behaviour

Factor	Finding
Age	Revenue is concentrated in the 20–29 and 30–39 age brackets (approximately 9M each), then declines sharply after age 40, with negligible contribution at 60+
Gender	The dashboard shows a balanced split with no clear single-gender dominance
Payment Method	Credit Card 41.64%, Debit Card 23.66%, Digital Wallet 19.25%, Bank Transfer 10.61%, Cash on Delivery 4.84%
Device	Mobile is the dominant shopping platform
Session Duration / Pages Viewed / Engagement Score	Approximately normally distributed, indicating relatively consistent browsing behavior across customers

E. Delivery and Satisfaction

Factor	Finding
Delayed Orders	32.24% of all orders
On-Time Orders	67.76%
Rating Distribution	Heavily concentrated toward 4-star and 5-star ratings; average rating is 3.90

Delivery Time Distribution	The majority of deliveries fall within the expected range, with a smaller tail of significant delays
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4. Analytical Insights and Observations

- Electronics and Bengaluru form the core growth engine.** Electronics contributes more than half of total revenue and is also the top-profit category, while Bengaluru generates roughly double the revenue of the next-best city. This concentration is currently a strength.
- Q1 represents a major seasonal spike rather than normal operating performance.** The rest of the year performs at roughly one-third of Q1's monthly pace, making seasonality the most significant trend in the dataset.
- Monetary variables are strongly right-skewed.** Sales, profit, unit price, and cost price are influenced by a relatively small number of high-value transactions, meaning average-based metrics overstate the typical customer profile.
- Customer browsing behaviour is consistent and predictable.** Session duration, pages viewed, and engagement score follow approximately normal distributions, with no erratic segment standing out.
- Moderate engagement appears to convert better than maximum engagement.** Customers who browse just enough tend to convert more effectively than those who browse extensively, suggesting that prolonged browsing may indicate hesitation rather than intent.
- Discounts do not independently guarantee profitability.** Profitability appears to be driven more by category and product mix than by discount depth alone.
- Product demand is balanced in volume, but not in margin.** Although unit demand is distributed across categories, Electronics remains the leading contributor to profitability.
- The customer base is digitally mature and mobile-first.** Credit cards are the leading payment method, cash on delivery is minimal, and mobile dominates device usage, but mobile-specific conversion tracking has not yet been validated.
- Ratings are positive but insufficiently diagnostic.** Because most ratings cluster at the high end, the scale offers limited sensitivity for identifying emerging service friction.
- Delivery delays affect a substantial minority of orders.** Nearly one-third of orders are delayed, which is significant enough to influence overall customer experience despite generally acceptable delivery-time distribution.
- Retention appears consistent across product categories.** The central challenge is not category-specific retention weakness, but re-engagement after the Q1 peak period.
- Smaller cities represent a meaningful but underdeveloped growth opportunity.** Their lower revenue contribution suggests untapped potential beyond the current metro-led hierarchy.

5. Problems Identified

#	Problem	Underlying Cause
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1	Revenue overly dependent on Q1	The exact driver is unclear; it may reflect seasonality, a one-time campaign, or front-loaded customer acquisition without a supporting retention strategy.
2	Heavy reliance on Electronics and Bengaluru	Both revenue and profit are concentrated in a single category and a single city, creating vulnerability if either weakens.
3	Revenue metrics distorted by a small number of high-value transactions	Strong right-skew in monetary variables causes average-based KPIs to overstate typical customer value.
4	Ratings system lacks diagnostic sensitivity	Ratings cluster too heavily at the high end to provide early visibility into emerging friction.
5	Discount spend not clearly tied to profitability	Broad or untargeted discounting likely includes customers who would have purchased anyway, creating discount leakage.
6	Category demand is balanced, but category profitability is not	Some categories may be consuming shelf space and operating resources without generating meaningful margin.
7	Mobile-first customer base without validated mobile experience data	No device-level conversion funnel tracking is currently in place.
8	High-engagement sessions may indicate hesitation, not intent	This pattern has not yet been investigated at the category or price-point level.
9	Nearly a third of orders arrive late	A logistics bottleneck likely exists and may be concentrated in specific cities or product categories.
10	Smaller cities remain underdeveloped	It is not yet clear whether low performance reflects a demand gap or an investment and infrastructure gap.

6. Recommended Actions

#	Problem Area	Recommended Action
1	Q1 revenue dependency	Identify the exact Q1 driver by cross-referencing orders with new-versus-returning customer status and any promotional flags. Build a structured Q2–Q4 reactivation campaign , including 30/60/90-day win-back emails or discounts targeted at customers acquired during Q1. Explore counter-seasonal category pushes to smooth the annual revenue curve.
2	Electronics and Bengaluru concentration	Track these two areas as core dependencies with dedicated monitoring and treat any decline as an early-

		warning signal. At the same time, invest in a second category and a second city as deliberate diversification priorities.
3	Skewed revenue distorting averages	Report median average order value alongside the mean. Build a customer value-tiering system, such as identifying the top 10% by spend, and monitor top accounts individually so that the loss of even one triggers an alert.
4	Ratings lack sensitivity	Add a brief post-delivery micro-survey, for example asking whether the delivery was on time, to capture issues that a 5-star scale may hide. Segment ratings by delivery status rather than relying only on the aggregate view.
5	Discount inefficiency	Conduct an incremental lift analysis comparing profit margins on discounted versus non-discounted orders within the same category and customer segment. Shift from broad category-wide discounting to targeted, behavior-triggered offers such as cart abandonment or first-time buyer promotions.
6	Demand-profit mismatch across categories	Perform a category-level profit margin audit rather than focusing only on revenue and volume. Determine whether low-margin categories are being maintained intentionally for cross-sell or acquisition purposes, or whether they require repricing or supplier renegotiation.
7	Unvalidated mobile experience	Add device-level funnel tracking to the dashboard using Device Type, Session Duration, Cart Size, and Completion Rate. Audit the mobile checkout flow in particular, as it is a common source of unobserved conversion loss.
8	High engagement without conversion	Investigate whether high-engagement, low-conversion sessions cluster around particular categories or price points, which may indicate price sensitivity or decision fatigue. Test simpler comparison tools or recommendation prompts for these sessions.
9	Delivery delays (32%)	Cross-reference delayed orders by city and product category to locate the main bottlenecks. Prioritize the top two to three delay-driving city or category combinations first, and track the direct relationship between delay and rating decline.
10	Underdeveloped smaller cities	Run a small-scale pilot in two to three tier-2 cities with dedicated marketing support, while checking delivery performance in those markets first. If logistics are already

		weaker there, that may itself be suppressing demand independent of marketing investment.
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7. Conclusion

This report indicates that the business has a strong commercial base, but its long-term resilience depends on reducing concentration risk, improving operational reliability, and sharpening performance measurement. The recommended priority actions are to **understand and address the Q1 drop, fix delivery delays, audit discount efficiency, build customer value tiers, and pilot smaller-city expansion after stabilization**. These actions are directly supported by the existing data and provide a practical path to more balanced and sustainable growth.